

Response to Reviewer #1

Manuscript: *Modeling oxygen-void interactions in uranium nitride*

We thank the reviewer for the constructive assessment. In this revision, we have carried out the additional calculations requested rather than relying on formal or literature arguments. All new material is cross-referenced below.

Part 1: Point-by-point response

1. DFT+U sensitivity declined ... a static-energy sensitivity test on the four energies that enter $\Delta\sigma$ (E_{inc} , E_{ad} , $E_{f,\text{bulk}}$, $E_{f,\text{surf}}$). B_0 and ΔH_f from Q1 are also missing; only a_0 is reported.

Response: We now report a static DFT+ U sensitivity test of the bulk energies that set the dominant $\text{O}_\text{N}^{(\text{s})}$ channel— $E_{\text{inc}}(\text{O}_\text{N}^{(\text{b})})$, $E_{\text{inc}}(\text{O}_i^{(\text{s})})$, and $E_b(\{\text{O}_\text{N} : v_\text{N}\})$ —at $U_{\text{eff}} = 1$ eV (new table in Appendix F). The oxygen-defect ordering is preserved: O_N remains more stable than O_i by 4.269 eV (versus 4.271 eV in PBE) and the binding energy stays near zero ($+0.027 \rightarrow -0.010$ eV). We attempted $U_{\text{eff}} = 2$ eV, but the static nitrogen-vacancy formation energy $E_f(v_\text{N}^{(\text{s})})$ came out *negative* (unphysical), indicating that a static calculation is inadequate at that U_{eff} (it would require occupation-matrix control or U -ramping); we therefore restrict the quantitative test to $U_{\text{eff}} = 1$ eV and state this explicitly. We further note that the dominant magnitude of $\Delta\sigma$ is set by the segregation energy $E_{\text{seg}} = -[E_f(v_\text{N}^{(\text{s})}) - E_f(v_\text{N}^{(\text{b})}) + E_{\text{ad}}(\text{O}_\text{N}^{(\text{s})}) - E_{\text{inc}}(\text{O}_\text{N}^{(\text{b})})]$. Because E_{seg} is a difference of surface and bulk energies, a systematic shift induced by U_{eff} acts in the same direction on both terms and largely cancels. Consequently, E_{seg} —and hence $\Delta\sigma$ —is substantially less sensitive to the choice of U_{eff} than any of the individual energies entering it, as quantified in Appendix F.

Finally, we now report $B_0 = 219.9$ GPa, $B'_0 = 4.77$, and $\Delta H_f(\text{UN}) = -2.697$ eV/f.u. (-260.18 kJ/mol-f.u.) in the Methodology section, in addition to a_0 .

2. Explicit small-void calculation declined ... the curvature/coordination dependence of $E_{\text{ad}}(\text{O}_\text{N}^{(\text{s})})$ and $E_f(v_\text{N}^{(\text{s})})$, which enter $\Delta\sigma$ linearly. A single nanovoid or stepped-surface calculation would resolve this.

Response: We have carried out an explicit slab calculation on the (110) facet, which directly probes the coordination sensitivity of the two surface energies the reviewer identifies. A new appendix (i.e., Appendix E) has been included to detail this calculation and relevant results and discussion. The (110) surface exposes an under-coordinated nitrogen sublattice, and both quantities change substantially relative to (001): $E_{\text{ad}}(\text{O}_\text{N}^{(\text{s})})$ weakens from -6.80 to -6.31 eV, while $E_f(v_\text{N}^{(\text{s})})$ drops sharply (surface segregation energy -1.57 eV versus -0.76 eV on (001)). Despite these large changes, the net $\text{O}_\text{N}^{(\text{s})}$ channel lowers the void surface energy at least as effectively as on (001). This bounds the coordination dependence of the two energies that enter $\Delta\sigma$ linearly and, together with the exact cancellation of the σ prefactor (retained in the Discussion), supports the planar-to-nanovoid extrapolation.

3. No original electronic-structure analysis ... the bonding picture should be verified, not assumed transferable.

Response: We have added an original charge-density-difference (CDD) analysis computed on our own $\text{O}_\text{N}^{(\text{s})}$ and $\text{O}_i^{(\text{s})}$ supercells with the same PBE workflow (new Section 3.3, with CDD maps and an O–U charge-density-metrics table). This independently reproduces the O 2p–U 5f interaction and charge transfer toward oxygen at the nitrogen-sublattice site, and explains the adsorption-energy ordering as a coordination-saturation effect, rather than assuming transferability from the cited works.

4. Faceting not quantified ... a single (110) or (111) slab calculation ... would substantiate it.

Response: The (110) slab calculation is presented in the new Appendix E and is now referenced in the Discussion (faceting paragraph), replacing the previous “beyond the scope” statement. It substantiates the local-coordination argument quantitatively.

5. Insufficient void-size sampling ... the simplifying assumptions of Appendix C should be validated against a denser grid of full-model evaluations.

Response: The comparison table in Appendix C now spans six discrete void radii ($R_v = 1, 2, 5, 10, 25, 50$ nm) and reports the corresponding $\Delta\sigma_\text{min}$ alongside the closed-form and full-model T_peak . The closed-form Appendix C result agrees with the full-model maximization to within 1 K across the entire grid, validating its simplifying assumptions. This table is now also referenced from the main text after Figure 5.

6. Planar approximation missing from abstract ... one sentence in the abstract would resolve it.

Response: The abstract now states: “As an approximation, surface energetics are obtained from planar (001) slab calculations and applied to curved void surfaces.”

7. t -uncertainty not propagated to the headline claim ... the “~58% nucleation-barrier reduction” carries roughly ± 25 percentage-point uncertainty.

Response: We have propagated the t -uncertainty at the point where the barrier reduction is reported. We note a clarification: the $\sim \pm 50\%$ variation of $|\Delta\sigma|$ applies to the kinetically limited *low-temperature flank*, whereas the barrier reduction is evaluated at T_peak , where $\alpha_1 = 1$ and $|\Delta\sigma|_\text{peak}$ varies by only -11% to $+7\%$ over $t = 10\text{--}100$ h. The barrier reduction is therefore 57–65%, and the full correlated DFT-plus- t propagation (Appendix F) gives a 95% interval of 45–78%. Both figures now appear where the result is presented.

8. No uncertainty band on Figure 6 ... propagating the quantified t -uncertainty and representative DFT-energy uncertainties would yield a defensible band.

Response: A correlated Monte Carlo uncertainty band, propagating ± 0.1 eV DFT-energy uncertainties (with shared systematic shifts for chemically related pairs) and the 10–100 h nucleation-time range, is now shown in the new figure in Appendix F, with the summary statistics tabulated. No sample violates $\sigma_\text{eff} > 0$.

Part 2: Additional edits made by the authors

While preparing this revision, we also identified and fixed the following:

- **Diffusivity figure.** We corrected a typo in the script that generated the oxygen-diffusivity plot. The correction did not alter the qualitative conclusions.
- **Capture length.** We refined the effective capture length to $\lambda_v = 0.5 n_v^{-1/3} - R_v$, subtracting the void radius from the half-spacing. This is more physically accurate for larger voids, where R_v is a non-negligible fraction of the void spacing. It shifted the quantitative results for the largest voids but left the qualitative conclusions unchanged.